The Matatu as a Natural Experiment

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Abstract

The matatu industry is a textbook, unregulated, high-friction implementation of the consolidation and yield problems the theory describes—which is why it sometimes looks chaotic and sometimes looks eerily efficient. This note maps matatu behaviour point by point onto the theory: waiting to fill is temporal consolidation executed manually, with a break-even load factor of 63.2% governed by $\lambda^* = c_0/[K(p - c_v)]$; route formation is spatial and demand-side consolidation that evolved spontaneously; the zero-margin equilibrium is the consolidation constant running without a kill rule; capacity choice is matched to density; and the December fare spike is primitive yield management. The deepest finding is what the industry lacks: a ledger. Every day starts from zero, so the sector is efficient at the individual departure and inefficient at the corridor, the network, and the decade.

Your theory predicts matatu behaviour with uncomfortable precision. The industry is a textbook, unregulated, high-friction implementation of the exact same consolidation and yield problems—which is why it sometimes looks chaotic and sometimes looks eerily efficient.

Here is the mapping, point by point.

1 "Waiting to Fill" Is Temporal Consolidation, Executed Manually

The single most recognisable matatu behaviour—sitting at the stage until full—is the temporal consolidation lever in its purest, crudest form.

The operator holds the departure until the vehicle crosses a load-factor threshold. That threshold is not arbitrary. A Kenyan PSV profitability calculator puts the break-even load factor for a 14-seater at 63.2%—below roughly nine passengers, the vehicle loses money that day regardless of how many trips it makes. That is λ^* from the consolidation constant,

$$\lambda^* = \frac{c_0}{K(p - c_v)}, \tag{1}$$

measured in the field.

The difference from your model is that the matatu has no batching window optimisation. It waits until full because waiting is the only consolidation mechanism it has. There is no yield-management layer to decide whether a longer wait is worth the marginal passenger, so the wait becomes socially costly—passengers sit, stages clog, and the vehicle eventually leaves when the conductor judges the load is "enough." It is the consolidation frontier operated by intuition rather than by the marginal condition.

The airline comparison in the theory is exact: the matatu operators themselves invoke it. The Federation of Public Transport argues that public transport cannot be a free market because "you can't allow planes to leave Wilson Airport at will, land at will, and take off at will, wait to fill passengers, and go". They are describing the absence of schedule discipline—the temporal consolidation lever deployed without coordination.

2 Route Formation Is Spatial and Demand-Side Consolidation, Evolved Spontaneously

Jarrett Walker's observation is the key: matatus "have organized themselves into routes because that's to their benefit; they train customers where to wait for them along reasonable paths so that they aren't driving around looking for customers individually".

That is spatial consolidation (a fixed route aggregates demand along a corridor) and demand-side consolidation (staging trains passengers to wait at known points rather than forcing the vehicle to search). It is exactly the logic of the bucket system—define a corridor, let demand accumulate at access points, dispatch when load is sufficient—except it evolved without a central designer.

The failure mode is also the one the theory predicts: every matatu wants to go downtown because downtown is the biggest market, so the network over-serves the radial corridor and under-serves crosstown demand. This is not stupidity. It is each operator individually optimising for the easiest place to fill a vehicle. The geometry of private consolidation produces a star-shaped network because that is what individual load-factor maximisation selects for.

3 The Zero-Margin Equilibrium Is the Consolidation Constant in Action

The medium piece on Nairobi transit states the dynamic explicitly: when a corridor becomes profitable, supply increases until "load factors fall, revenue per vehicle falls, and effective margins collapse".

That is the consolidation constant operating without a kill rule. New vehicles enter a profitable corridor, each one raises total capacity, each one lowers the average load factor, and the corridor converges to zero margin. The industry is stuck in a low-level equilibrium because there is no mechanism to kill under-loaded departures—an operator who leaves with six passengers still earns something on the day, and the daily cash target makes "something" better than nothing.

This is precisely why the bucket system's kill threshold matters. A matatu with no kill rule dispatches at any load above zero, and the aggregate effect is the erosion of corridor margin. The industry's own proposal—gross-cost contracts where the operator is paid per kilometre regardless of revenue—is an attempt to remove the load-factor constraint from the operator and place it on a contracting authority. That is a recognition, in policy language, that the consolidation problem cannot be solved by the operator alone.

4 Capacity Choice Is Matched to Density—Badly, but It Is Matched

The "baby matatu" phenomenon—smaller vehicles that fill faster—is the capacity-choice proposition made visible. A 14-seater needs roughly nine passengers to break even; a smaller vehicle needs fewer. In low-density corridors, the smaller vehicle is the correct K^* because it can actually fill. The Standard reports that baby matatus "carry a few passengers and take a shorter time to fill up," and that their price moderates the bus fare through competition.

The theory says the same thing: choose K for the corridor's density, not for the ambition. The matatu industry has discovered this through trial and error. What it has not discovered is the diagnostic—the utilisation ratio that would tell an operator whether a corridor is consolidation-constrained (needs smaller vehicles or longer batching) or addressability-constrained (needs better staging and route knowledge).

5 The December Fare Spike Is Primitive Yield Management

Matatu fares spike in December because demand rises and capacity is fixed in the short run. The Star notes that "unlike airlines, which publish seasonal fare calendars months in advance, the matatu industry responds to demand almost instantly," and that operators treat Christmas as a "peak harvest period" to offset months of thin margins.

That is peak-load pricing—the temporal and demand-side levers applied through price rather than through scheduling. It is yield management, but without the yield-management apparatus. There is no advance-purchase discount, no service-level menu, no capacity inventory. The fare moves because the conductor can see the queue. The result is effective in the short run and destructive in the long run, because it trains passengers to expect arbitrary pricing and erodes trust in the service.

6 Why Matatus Cannot Escape the Trap: The Missing Ledger and the Daily Cash Target

The deepest connection is the one the theory identifies as the difference between a bootstrap and a moat: the matatu industry has no ledger.

The financial model report is blunt: the industry runs on "daily cash targets rather than profitability through efficiency improvements and operational optimization". Loans are amortised over as little as 1.5 years, which forces the driver to generate revenue immediately rather than build a durable operating advantage. There is no mechanism to record which stages are reliably productive, which times of day have predictable demand, or which routes compound over time. Every day starts from zero.

In the language of the flywheel, the matatu industry is running an open-loop system. It delivers passengers, but it does not convert those deliveries into data that would raise its own future efficiency. It cannot accumulate addressability—not of places, but of demand patterns—because it does not record the patterns in a form that compounds. The result is a sector that is remarkably efficient at the level of the individual departure and remarkably inefficient at the level of the corridor, the network, and the decade.

The industry's own diagnosis points at the same gap. The Federation of Public Transport is asking for route planning, fleet limits, and long-term contracts precisely because the individual operator cannot solve the consolidation problem alone. That is the correct diagnosis. What the theory adds is the reason: consolidation requires a ledger, and the ledger requires an institution willing to hold it.

7 The Matatu as a Natural Experiment

The matatu industry is what the consolidation and yield problems look like when they are solved by evolution rather than by design. It gets the local answers right—routes form, vehicles fill, capacity adapts, fares respond to demand—and the global answers wrong—corridors oversupply, margins collapse, networks starve crosstown demand, and the sector cannot reinvest.

That is the strongest possible validation of the theory. The laws are operating. The industry is obeying them. What it lacks is the diagnostic—the three numbers (drops per vehicle-hour, utilisation ratio, kill rate) that would tell an operator or a regulator which lever to pull, and the ledger that would let the effect of pulling it compound.

If you put a bucket system with a kill rule and a demand ledger into a matatu SACCO, you would not be imposing an alien model. You would be giving the industry the measurement layer it has never had, to solve a problem it has been solving by trial and error for sixty years.